

Statistics for Machine Learning — Part 1

A placement-focused guide to descriptive and inferential statistics. In every ML workflow, compute statistics on training data first; reuse those fitted values on validation and test data to avoid leakage.

Mean, Median, and Mode

1. Overview

Mean, median, and mode are measures of central tendency: compact answers to “what is typical in this feature?” They are used in EDA, data-quality reports, missing-value imputation, baseline models, and feature monitoring.

2. Intuition

For salaries `[30, 35, 40, 45, 300]`, the mean is pulled toward the CEO’s salary, while the median better describes a typical employee. Mode answers which value or category appears most often.

3. Prerequisites

Arithmetic; sorting; numeric versus categorical variables; NumPy/Pandas arrays.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Mean	Sum of all values divided by count	Uses every magnitude but is sensitive to outliers.
Median	Middle value after sorting	Robust center; average the two middle values when <code>n</code> is even.
Mode	Most frequent value	Works for categorical data; may be absent or multiple.
Weighted mean	$\sum w_i x_i / \sum w_i$	Used when samples have different importance.

5. Algorithm / Working Process

Input one feature column. Sum and divide for the mean; sort and choose the middle for the median; count values and choose the highest count for the mode. Output one representative value, except multimodal data can return several modes.

6. Mathematical Foundation

For $x_1, \dots, x_n$, $\bar{x} = (1/n) \sum x_i$. Median is the 50th percentile. Mode is `argmax_v count(x_i=v)`. The mean minimizes total squared error; the median minimizes total absolute error.

7. Practical Implementation

```
import numpy as np
from collections import Counter

x = np.array([30, 35, 40, 45, 300])
print(x.mean(), np.median(x), Counter(x).most_common(1)[0][0])

# Fit this statistic on train data only in a real pipeline.
x_nan = np.array([30., 35., np.nan, 45., 300.])
filled = np.where(np.isnan(x_nan), np.nanmedian(x_nan), x_nan)
print(filled)
```

8. Code Explanation

NumPy computes vectorized mean and median. `Counter` counts discrete values. `nanmedian` ignores missing values, and `where` replaces only missing entries.

9. Training / Evaluation

These are preprocessing statistics, not trained models. Fit an imputer on training rows only. Median is often a strong regression baseline for MAE; mean is the equivalent baseline for MSE.

10. Complexity and Cost

Mean is $O(n)$ time and constant extra memory. Median is commonly $O(n \log n)$ when sorted; mode is $O(n)$ time and $O(k)$ memory for k unique values. CPU is enough.

11. Common Use Cases

EDA summaries; median imputation for skewed columns; most-common-category imputation; monitoring feature center; regression baselines.

12. Common Mistakes

Using mean for heavy-tailed income; averaging arbitrary category IDs; computing imputation before a split; assuming mode is always unique.

13. Edge Cases / Limitations

Mean is outlier-sensitive; median loses magnitude detail; mode is unstable for continuous values. All three can conceal multimodal distributions.

14. Variations

Trimmed mean removes tails; geometric mean handles multiplicative growth; harmonic mean handles rates; weighted mean handles importance. Mean/median/mode are placement essentials.

15. Related Topics

Quantiles generalize median. Skewness explains mean–median gaps. MSE targets the mean; MAE targets the median.

16. Interview Questions

1. **Mean vs median?** Mean uses all magnitudes; median resists outliers.
2. **When use mode?** For most frequent category or repeated discrete value.
3. **Can mode be multiple?** Yes, a dataset may be multimodal.
4. **Median with even count?** Average the two sorted middle values.
5. **Which minimizes MSE?** Mean.
6. **Which minimizes MAE?** Median.
7. **Why can mean mislead?** One extreme value shifts it substantially.
8. **Can categorical codes be averaged?** Usually no; encoding numbers are arbitrary.
9. **What does mean > median often suggest?** Right skew.
10. **Income imputation choice?** Usually median after distribution inspection.

17. Practice Tasks

Implement all three without libraries; add outliers and compare summaries; perform train-only median imputation; plot a histogram where summaries hide two clusters.

18. Project Ideas

Project	Stack and dataset	Resume value
Data profiler	Pandas, Streamlit; arbitrary CSV	Automated robust EDA report.
Salary explorer	Pandas, Plotly; salary survey	Explains skew and representative center.
Drift monitor	Python, Evidently; event logs	Detects changes in feature center.

19. Quick Revision

Key idea: summarize center. Main formula: $\sum x/n$. Use median with outliers. Trap: mean is not the most common value. Interview one-liner: mean is efficient; median is robust.

20. Final Cheat Sheet

Definition: center summaries. **Input/output:** column → representative value. **Steps:** aggregate, sort, or count. **Metrics:** compare with quantiles/plots. **Pros:** simple. **Cons:** information loss. **Best use:** EDA and train-only imputation.

Variance and Standard Deviation

1. Overview

Variance and standard deviation measure spread around the mean. They support feature scaling, anomaly detection, target-noise analysis, uncertainty reasoning, and normalization in ML pipelines.

2. Intuition

Two classes may average 70 marks, but one ranges 68–72 and another 20–100. The second has larger variance. Standard deviation expresses spread in the original unit, unlike variance.

3. Prerequisites

Mean, deviations, squares, square roots, basic probability, and z-score normalization.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Deviation	$x_i - \bar{x}$	Signed deviations sum to zero.
Variance	Average squared deviation	Penalizes large departures.
Standard deviation	Square root of variance	Has the original unit and is interpretable.
Sample vs population	Divide by $n-1$ vs N	Explain Bessel's correction and ddof .

5. Algorithm / Working Process

Compute mean, subtract it from each observation, square deviations, average them, then square-root for standard deviation. Input is a numeric feature; output is non-negative spread.

6. Mathematical Foundation

Population: $\sigma^2 = (1/N) \sum (x_i - \mu)^2$, $\sigma = \sqrt{\sigma^2}$. Sample variance: $s^2 = \sum (x_i - \bar{x})^2 / (n-1)$. Z-score: $z = (x - \bar{x}) / s$.

7. Practical Implementation

```
import numpy as np

x = np.array([2., 4., 4., 4., 5., 5., 7., 9.])
print("population", np.var(x), np.std(x))
print("sample variance", np.var(x, ddof=1))
z = (x - x.mean()) / x.std()
assert np.isclose(z.mean(), 0) and np.isclose(z.std(), 1)
```

8. Code Explanation

NumPy defaults to population calculations (**ddof=0**). **ddof=1** uses the sample denominator. The assertion is a minimal check that z-score scaling centered and rescaled this sample.

9. Training / Evaluation

Use a training-fitted scaler for KNN, SVM, linear models, PCA, and neural networks. Tree models generally do not need scaling. Judge scaling by held-out model metrics, not only by unit variance.

10. Complexity and Cost

Time is $O(n)$; streaming algorithms can use constant extra memory. Avoid numerically fragile manual formulas for huge values. CPU is normally sufficient.

11. Common Use Cases

Standardization; control limits; anomaly detection; residual diagnostics; volatility; batch-normalization intuition.

12. Common Mistakes

Confusing variance with standard deviation; mixing n and $n-1$; fitting scaler before split; treating standard deviations as universal normal-distribution guarantees.

13. Edge Cases / Limitations

Outliers inflate both. A constant column has zero standard deviation and cannot be z-scored without a guard. IQR/MAD are better for heavy-tailed data.

14. Variations

Weighted, running (Welford), robust (MAD/IQR), and conditional variance. Standard deviation and standardization are interview priorities.

15. Related Topics

Variance is covariance with itself. PCA maximizes projected variance. Bias–variance trade-off includes model prediction variance.

16. Interview Questions

1. **Why square deviations?** To prevent cancellation and weight large errors.
2. **Variance vs standard deviation?** Squared units versus original units.
3. **Why $n-1$?** It corrects sample underestimation of population variance.
4. **Can variance be negative?** No.
5. **What does zero variance mean?** Every value is identical.
6. **Effect of multiplying data by 3?** Variance multiplies by 9.
7. **Why scale before KNN?** Large units dominate distances.
8. **Do decision trees need scaling?** Usually no.
9. **Robust spread alternative?** IQR or MAD.
10. **High target variance suggests?** A noisier, harder prediction problem.

17. Practice Tasks

Calculate variance manually; compare both $ddof$ choices; scale train/test safely; show one outlier's impact on standard deviation and IQR.

18. Project Ideas

Project	Stack and dataset	Resume value
Sensor anomaly alerts	Pandas, NumPy; IoT data	Statistical monitoring baseline.
Scaling benchmark	scikit-learn; UCI data	Measures impact on KNN/SVM.
Streaming variance monitor	Python; simulated events	Shows online-data competence.

19. Quick Revision

Key idea: spread. Formula: average squared deviation. Use std in original units. Trap: fitting scaling on all data. One-liner: variance quantifies dispersion and drives normalization.

20. Final Cheat Sheet

Input/output: numbers → spread. **Steps:** mean, deviations, square, average, root. **Key choice:** `ddof`. **Pros:** fundamental. **Cons:** outlier-sensitive. **Best use:** scaling and data quality.

Correlation

1. Overview

Correlation quantifies strength and direction of association. It helps with EDA, feature redundancy, multicollinearity checks, and monitoring relationships, but does not establish causality.

2. Intuition

Study hours and marks may move upward together (positive); price and demand may move oppositely (negative). A zero Pearson correlation means no linear relationship—not no relationship at all.

3. Prerequisites

Mean, standard deviation, covariance, scatter plots, and paired observations.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Pearson r	Linear association in $[-1, 1]$	Interpret sign, magnitude, and causal limitation.
Spearman ρ	Pearson correlation of ranks	Use for ordinal/monotonic trends or outliers.
Correlation matrix	Pairwise values across features	Finds redundancy and suspicious leakage.
Spurious correlation	Coincidence or confounding	Correlation alone never proves mechanism.

5. Algorithm / Working Process

Align paired numeric rows, handle missing values consistently, inspect a scatter plot, calculate a suitable coefficient, then assess whether the relationship is useful on held-out data.

6. Mathematical Foundation

$r = \text{cov}(X, Y) / (s_X s_Y) = \frac{\sum((x_i - \bar{x})(y_i - \bar{y}))}{\sqrt{(\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2)}}$. It is undefined if either variable has zero standard deviation.

7. Practical Implementation

```
import pandas as pd
from scipy.stats import pearsonr, spearmanr

df = pd.DataFrame({"hours": [1, 2, 3, 4, 5], "score": [45, 50, 65, 70, 85]})
r, p = pearsonr(df.hours, df.score)
rho, _ = spearmanr(df.hours, df.score)
print(f"Pearson r={r:.3f}, p={p:.4f}; Spearman rho={rho:.3f}")
print(df.corr(numeric_only=True))
```

8. Code Explanation

`pearsonr` returns coefficient plus a zero-correlation p-value under its assumptions. Spearman ranks values first. A DataFrame correlation matrix scales pairwise inspection to many numeric columns.

9. Training / Evaluation

Calculate correlation-based feature filters on training data only. Validate model value on a held-out set: a feature can correlate with the target but add nothing after other features, or be leakage.

10. Complexity and Cost

One pair is $O(n)$. A dense matrix for d features is about $O(nd^2)$ time and $O(d^2)$ memory. CPU is sufficient for ordinary tabular data.

11. Common Use Cases

EDA heatmaps; redundant-feature screening; linear-regression multicollinearity diagnosis; financial co-movement; feature-target monitoring.

12. Common Mistakes

Claiming causation; using Pearson for a curved relationship without plotting; ignoring outliers; blindly removing correlated features from trees; leaking target-derived features.

13. Edge Cases / Limitations

Pearson misses U-shaped dependence and is outlier-sensitive. Range restriction changes it. Zero correlation does not generally mean independence.

14. Variations

Spearman, Kendall tau, partial correlation, point-biserial correlation, and autocorrelation. Pearson/Spearman are key placement topics.

15. Related Topics

Covariance is unnormalized correlation. VIF measures multicollinearity. Mutual information detects wider nonlinear dependence. Causal inference asks why association exists.

16. Interview Questions

1. **Pearson range?** $[-1, 1]$.
2. **Does $r=0$ imply independence?** No, only no linear association.
3. **Does $r=1$ prove causation?** No.
4. **Why is r unitless?** It divides covariance by both standard deviations.
5. **When use Spearman?** Ordinal or monotonic/outlier-prone data.
6. **When undefined?** A constant variable.
7. **Can high correlation be leakage?** Yes; inspect data origin and time.
8. **What is multicollinearity?** Predictors strongly correlate with each other.
9. **Do trees require uncorrelated inputs?** No.
10. **Best inspection method?** Coefficient plus scatter plot and domain context.

17. Practice Tasks

Simulate linear, U-shaped, and outlier data; compare Pearson/Spearman; build a heatmap; find a leaked feature in a toy pipeline.

18. Project Ideas

Project	Stack and dataset	Resume value
Correlation auditor	Pandas, Plotly	Flags collinearity and leakage risks.
Asset relationship explorer	Pandas, Plotly; returns	Studies changing correlations.
Feature filter study	sklearn; UCI data	Validates EDA choices on holdout data.

19. Quick Revision

Key idea: standardized linear association. Formula: $\text{cov}/(\text{std} \times \text{std})$. Use a plot. Trap: causation claims. One-liner: correlation is an EDA signal, not causal proof.

20. Final Cheat Sheet

Input/output: paired columns $\rightarrow [-1, 1]$. **Steps:** align, inspect, compute, validate. **Metrics:** r , ρ , CI/p-value. **Pros:** interpretable. **Cons:** linear/outlier limitations. **Best use:** EDA.

Covariance

1. Overview

Covariance measures whether variables deviate from their means together. It powers covariance matrices, PCA, multivariate Gaussian models, Mahalanobis distance, and risk analysis.

2. Intuition

If people who are above-average in height are usually also above-average in weight, products of their deviations are positive, so covariance is positive.

3. Prerequisites

Mean, variance, deviations, matrix multiplication, and paired observations.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Sign	Positive together; negative opposite	Indicates joint direction, not comparable magnitude.
Covariance matrix	Diagonal variances; off-diagonal covariances	Input to PCA and Gaussian methods.
Scale dependence	Units alter magnitude	Explains why correlation is often easier to compare.
PSD property	Valid matrix is symmetric positive semidefinite	Important for numerical validity.

5. Algorithm / Working Process

Center every feature by its mean. Multiply paired centered values and average for a scalar covariance. For many features, multiply the centered matrix transpose by itself and divide by the degrees-of-freedom denominator.

6. Mathematical Foundation

$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$. Sample covariance is $\sum (x_i - \bar{x})(y_i - \bar{y}) / (n-1)$. For centered rows-as-samples matrix X_c , $S = X_c^T X_c / (n-1)$.

7. Practical Implementation

```
import numpy as np

X = np.array([[1., 2.], [2., 4.], [3., 5.], [4., 8.]])
S = np.cov(X, rowvar=False, ddof=1)
print(S)
assert np.allclose(S, S.T)
print("joint covariance:", S[0, 1])
```

8. Code Explanation

`rowvar=False` says columns are variables. The diagonal is sample variance; off-diagonal entries measure joint spread. Symmetry is an essential covariance-matrix check.

9. Training / Evaluation

Estimate covariance on training data for PCA, Gaussian classifiers, or whitening. When features outnumber samples, regularize/shrink covariance and validate the downstream result.

10. Complexity and Cost

For n samples and d features, dense covariance costs roughly $O(nd^2)$ time and $O(d^2)$ memory. Large d often needs dimensionality reduction or shrinkage.

11. Common Use Cases

PCA; portfolio diversification; multivariate anomaly detection; Kalman filters; Gaussian discriminant analysis; whitening.

12. Common Mistakes

Using rows/features in the wrong orientation; comparing covariance across differently scaled units; forgetting centering; inverting a singular matrix; reading covariance as causation.

13. Edge Cases / Limitations

Outliers distort it. Categorical columns do not suit ordinary covariance. Small-sample high-dimensional estimates can be singular/noisy. Zero covariance is not general independence.

14. Variations

Population/sample, robust, shrinkage (Ledoit–Wolf), autocovariance, and cross-covariance. Covariance matrices and PCA are interview-relevant.

15. Related Topics

$\text{Var}(X) = \text{Cov}(X, X)$. Correlation normalizes covariance. PCA eigendecomposes it. Mahalanobis distance uses inverse covariance.

16. Interview Questions

1. **Variance relationship?** $\text{Var}(X) = \text{Cov}(X, X)$.
2. **Is covariance symmetric?** Yes.
3. **Why not compare magnitudes directly?** Units affect them.
4. **Diagonal of a covariance matrix?** Feature variances.
5. **Positive covariance means?** Values tend to move together.
6. **Does zero imply independence?** Not generally.
7. **Why divide by $n-1$?** Sample correction.
8. **PCA connection?** Components are variance-maximizing covariance directions.

9. **Why is singularity bad?** Matrix inverse is unavailable.
10. **Fix high dimension?** Shrinkage or reduce dimensions.

17. Practice Tasks

Calculate a 2×2 matrix by hand; verify diagonal variance; compare empirical/shrinkage covariance; apply PCA before and after scaling.

18. Project Ideas

Project	Stack and dataset	Resume value
PCA image compressor	NumPy, sklearn; Fashion-MNIST	Explains covariance-to-components pipeline.
Portfolio risk dashboard	Pandas, Plotly; asset returns	Demonstrates diversification math.
Multivariate anomaly detector	sklearn; sensor readings	Uses Mahalanobis distance.

19. Quick Revision

Key idea: joint spread. Formula: average product of deviations. Trap: scale-dependent values. One-liner: correlation is normalized covariance.

20. Final Cheat Sheet

Input/output: paired features $\rightarrow$ scalar/matrix. **Steps:** center, multiply, average. **Key choice:** sample vs population/shrinkage. **Pros:** multivariate foundation. **Cons:** scale and outlier sensitivity. **Best use:** PCA/Gaussian methods.

Sampling

1. Overview

Sampling chooses observations from a population so we can train, evaluate, and infer without measuring everything. It is foundational for datasets, minibatches, A/B tests, labeling, and reliable model evaluation.

2. Intuition

A city survey should sample residents broadly, not only friends. A representative random subset can estimate population behavior; a biased subset cannot be rescued merely by making it large.

3. Prerequisites

Population versus sample, probability, random seeds, class labels, and train/validation/test splitting.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Random sampling	Known selection chance	Reduces selection bias.
With replacement	Units can repeat	Bootstrap draws are independent.
Stratified sampling	Preserves group proportions	Essential for imbalanced classification splits.
Cluster/systematic	Sample groups/every kth unit	Practical but can inherit pattern bias.
Bootstrap	Resample observed sample	Estimates uncertainty without new data.

5. Algorithm / Working Process

Define target population and sampling frame; choose a design; sample with a reproducible seed; verify representation; isolate test data. For ML, fit preprocessing only after splitting and use minibatches during training.

6. Mathematical Foundation

The sample mean estimates population mean. Under IID sampling, standard error is approximately $s/\sqrt{n}$. Random error decreases at rate $1/\sqrt{n}$; systematic selection bias does not disappear with larger n .

7. Practical Implementation

```
import numpy as np
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

X, y = load_breast_cancer(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)
rng = np.random.default_rng(42)
means = [rng.choice(X_train[:, 0], len(X_train), replace=True).mean() for _ in
range(1000)]
print(y.mean(), y_test.mean(), np.percentile(means, [2.5, 97.5]))
```

8. Code Explanation

`stratify=y` preserves class balance approximately; `random_state` makes results reproducible. Each bootstrap draw resamples training rows with replacement; percentile bounds approximate a 95% interval for a mean.

9. Training / Evaluation

Use stratified splits for imbalanced labels, group splits for repeated users/patients, and chronological splits for time series. Never let duplicates, augmented variants, or future data cross into evaluation.

10. Complexity and Cost

Basic sampling is $O(n)$. B bootstrap draws cost $O(Bn)$. Minibatches reduce memory and make large-model optimization feasible.

11. Common Use Cases

Surveys; data labeling; train/test split; SGD minibatches; class-balanced batches; bootstrap confidence intervals; A/B experiments.

12. Common Mistakes

Convenience sampling; random time-series split; no stratification for rare labels; duplicate leakage; repeatedly using test data; unseeded experiments.

13. Edge Cases / Limitations

Missing groups in the sampling frame remain unobserved. Tiny rare classes may not split well. IID assumptions fail in time, geography, networks, and repeated-user data.

14. Variations

Simple random, stratified, cluster, systematic, reservoir, importance, and active sampling. Stratification and bootstrap are placement essentials.

15. Related Topics

Sampling distributions support CIs and tests. Minibatch sampling enables SGD. Cross-validation repeatedly resamples partitions.

16. Interview Questions

1. **Population vs sample?** Whole target group versus observed subset.
2. **Why stratify?** Preserve label proportions.
3. **With replacement?** A unit may occur repeatedly.
4. **Why bootstrap?** Estimate uncertainty from existing observations.
5. **Does bigger n remove bias?** No.
6. **Why not random split time data?** Future leakage.
7. **Sampling frame?** Source list/process from which units are selected.
8. **Why seed?** Reproducibility.
9. **How split grouped data?** A group-aware splitter.
10. **What is a minibatch?** Small sample used for one gradient estimate.

17. Practice Tasks

Compare random/stratified splits; bootstrap a metric; simulate sample means at several n ; show temporal leakage from a random split.

18. Project Ideas

Project	Stack and dataset	Resume value
Split auditor	sklearn, Pandas	Detects label/group/time imbalance.
Bootstrap metric reporter	sklearn, NumPy	Adds uncertainty to model metrics.
Active-labeling simulator	Python, sklearn; text/images	Shows data-efficient learning.

19. Quick Revision

Key idea: representative subset. Formula: $SE = s/\sqrt{n}$. Use stratification for imbalance. Trap: random split is not always valid. One-liner: more data reduces variance, not bias.

20. Final Cheat Sheet

Input/output: population/frame → sample/splits. **Steps:** define target, choose method, sample, audit. **Key choices:** seed, strata, replacement. **Pros:** scalable. **Cons:** selection bias. **Best use:** all empirical ML.

Bias and Variance

1. Overview

Bias and variance explain generalization error from overly simple assumptions versus excessive sensitivity to training data. They guide model capacity, regularization, ensembles, and debugging of underfitting/overfitting.

2. Intuition

A line forced through a curved pattern is consistently wrong (high bias). A deep tree that changes dramatically after a few rows change has high variance. Good generalization balances them.

3. Prerequisites

Supervised learning, train/validation error, expectation, squared loss, regularization, and cross-validation.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Bias	Difference between average prediction and true function	High bias produces underfitting.
Variance	Prediction variation across training sets	High variance produces overfitting.
Irreducible noise	Unobservable randomness in target	Cannot be removed by complexity.
Regularization	Complexity penalty	Typically raises bias and lowers variance.

5. Algorithm / Working Process

Train models of several capacities, compare train and cross-validation loss, then act on the pattern. High train and validation loss suggests bias; low train but high validation loss suggests variance.

6. Mathematical Foundation

For squared loss: $E[(Y - \hat{f}(X))^2] = \text{Bias}(\hat{f}(X))^2 + \text{Var}(\hat{f}(X)) + \sigma^2_{\text{noise}}$. Validation loss estimates the total, not perfect individual components from one dataset.

7. Practical Implementation

```
from sklearn.datasets import make_regression
from sklearn.model_selection import validation_curve
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import Ridge

X, y = make_regression(n_samples=200, n_features=1, noise=25, random_state=42)
model = make_pipeline(PolynomialFeatures(), Ridge())
degree, train, valid = validation_curve(
    model, X, y, param_name="polynomialfeatures__degree",
    param_range=range(1, 11), scoring="neg_mean_squared_error", cv=5
)
print(list(zip(degree, -train.mean(1), -valid.mean(1))))
```

8. Code Explanation

The validation curve trains increasing polynomial capacities under five folds. MSE is negated by sklearn's scoring convention. A widening train-validation gap indicates variance; uniformly bad scores indicate bias or weak features.

9. Training / Evaluation

Use cross-validation and reserve final test data. To reduce variance: get data, augment, regularize, prune, early-stop, use dropout/bagging. To reduce bias: improve features, train longer, reduce regularization, or increase appropriate capacity.

10. Complexity and Cost

Higher capacity and ensembles raise training, inference, and memory cost. Cross-validation multiplies training by the number of folds.

11. Common Use Cases

Choosing tree depth; neural-network regularization; learning-curve diagnosis; bagging; boosting; hyperparameter tuning.

12. Common Mistakes

Calling all poor accuracy overfitting; choosing by train score; tuning on test data; assuming complexity always helps; ignoring label noise or split mismatch.

13. Edge Cases / Limitations

The textbook decomposition is cleanest for regression squared loss. Dataset shift can dominate both. A validation gap may come from a bad split, not only model variance.

14. Variations

Learning curves, bagging (variance reduction), boosting (often bias reduction), Bayesian uncertainty, double descent. This trade-off is a core placement concept.

15. Related Topics

Overfitting/underfitting operationalize the trade-off. Regularization controls capacity. Cross-validation estimates generalization. Ensembles trade compute for stability.

16. Interview Questions

1. **High-bias pattern?** High train and validation loss.
2. **High-variance pattern?** Low train, high validation loss.
3. **What does more data help?** Usually variance.
4. **Regularization effect?** Higher bias, lower variance.
5. **Bagging effect?** Mainly lower variance.
6. **Boosting effect?** Often lower bias, may fit noise.
7. **Can test loss rise with capacity?** Yes.
8. **Irreducible error?** Noise unavailable in inputs.
9. **How diagnose?** Learning and validation curves.
10. **Can leakage hide variance?** Yes, with unrealistically good validation.

17. Practice Tasks

Plot polynomial curves; vary tree depth; compare a tree to Random Forest; add label noise; use learning curves to recommend a next fix.

18. Project Ideas

Project	Stack and dataset	Resume value
Diagnosis dashboard	sklearn, Streamlit	Turns error patterns into actions.
Ensemble benchmark	sklearn; California Housing	Quantifies bagging benefits.
Augmentation study	PyTorch; CIFAR-10	Demonstrates variance control.

19. Quick Revision

Key idea: systematic error versus instability. Formula: $\text{bias}^2 + \text{variance} + \text{noise}$. Trap: train score alone. One-liner: generalization needs the capacity sweet spot.

20. Final Cheat Sheet

Input/output: model/data → error diagnosis. **Steps:** correct split, curves, tune capacity. **Hyperparameters:** depth, alpha, dropout. **Metrics:** held-out loss. **Pros:** actionable. **Cons:** idealized decomposition. **Best use:** model debugging.

Hypothesis Testing

1. Overview

Hypothesis testing is a decision framework for assessing whether observed data is sufficiently inconsistent with a baseline claim. It is used in A/B tests, feature experiments, model comparisons, and scientific studies.

2. Intuition

A recommender's conversion rises from 10.0% to 10.3%. The test asks whether a difference this large would be unusual if the true lift were zero, rather than declaring success from the raw difference alone.

3. Prerequisites

Probability distributions, sampling, standard errors, null/alternative hypotheses, confidence intervals, and significance levels.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Null H_0	Baseline/no-effect claim	Tests seek evidence against it, not proof it is true.
Alternative H_1	Competing claim	Specify one- or two-sided direction before seeing data.
Type I/II error	False positive / false negative	α controls Type I; power addresses Type II.
Test statistic	Standardized discrepancy	Determines the null distribution and p-value.
Power	Detecting a true effect	Drives sample-size planning.

5. Algorithm / Working Process

State a measurable question and H_0/H_1 ; choose method/assumptions and α in advance; collect independent representative data; compute statistic, p-value, effect size, and CI; reject or fail to reject H_0 ; report practical impact.

6. Mathematical Foundation

One-sample t statistic: $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$. Compare it with a t distribution with $n-1$ degrees of freedom. For two independent means, Welch's t-test is a good default when variances may differ.

7. Practical Implementation

```
import numpy as np
from scipy.stats import ttest_ind

rng = np.random.default_rng(42)
control = rng.normal(100, 12, 200)
treatment = rng.normal(103, 12, 200)
t, p = ttest_ind(treatment, control, equal_var=False)
print(f"lift={treatment.mean()-control.mean():.2f}, t={t:.2f}, p={p:.4f}, reject={p < .05}")
```

8. Code Explanation

The code simulates independent groups. Welch's test avoids assuming equal variance. The result must include lift, not only p-value: a statistically detectable difference may be operationally trivial.

9. Training / Evaluation

For ML model comparisons on the same examples, preserve paired errors/predictions and use a paired method or paired bootstrap. Predeclare primary metric, stopping rule, guardrails, and minimum useful effect.

10. Complexity and Cost

Common analytic tests are $O(n)$; computation is cheap. Obtaining enough independent data or training repetitions for reliable power is usually expensive.

11. Common Use Cases

A/B tests; conversion comparison; feature impact; model-error comparison; scientific evaluation.

12. Common Mistakes

Peeking/stopping after significance; uncorrected multiple tests; post-hoc hypotheses; p-value-only reporting; confusing statistical and practical significance; biased instrumentation.

13. Edge Cases / Limitations

Huge samples make trivial effects significant. Small samples miss real effects. Dependence, contamination, nonstationarity, and poor sampling invalidate nominal conclusions.

14. Variations

z-test, Welch/paired t-test, chi-square/Fisher test, Mann–Whitney, permutation test, Bayesian A/B analysis. t-test and error types are placement priorities.

15. Related Topics

p-values quantify null-data compatibility. Confidence intervals show uncertainty. Sampling and power control reliability. Causal conclusions require good experimental design.

16. Interview Questions

1. **What is H_0 ?** Baseline/no-effect claim.
2. **Reject or accept?** Reject or fail to reject; never prove it true.
3. **Type I error?** False positive.
4. **Type II error?** Missing a real effect.
5. **Power?** Probability of detection when an effect exists.
6. **Why Welch?** It handles unequal variances.
7. **One vs two tailed?** Directional predeclared claim versus either direction.
8. **Practical significance?** Effect is large enough to matter.
9. **Why correct multiple tests?** Many tests inflate false positives.
10. **Can a test prove a model always wins?** No, it is conditional on sampled conditions.

17. Practice Tasks

Test two model-error arrays; simulate 1,000 null experiments; vary sample size while holding effect fixed; implement a permutation test.

18. Project Ideas

Project	Stack and dataset	Resume value
A/B analyzer	Streamlit, SciPy	End-to-end experiment reporting.
Model comparison lab	sklearn, SciPy; UCI data	Uses paired uncertainty analysis.
Power calculator	NumPy, statsmodels	Plans experiment sample size.

19. Quick Revision

Key idea: evidence against a baseline. Formula: $t = (\text{estimate} - \text{null}) / SE$. Use a predeclared test. Trap: p-value is not probability the null is true. One-liner: report effect, CI, and p-value together.

20. Final Cheat Sheet

Input/output: experiment $\rightarrow$ decision plus uncertainty. **Steps:** predefine, test, contextualize. **Key choices:** alternative, α , power. **Pros:** disciplined comparison. **Cons:** design-sensitive. **Best use:** A/B and model experiments.

p-value

1. Overview

A p-value is the probability, assuming the null hypothesis and test assumptions, of data at least as extreme as the observed result. It is evidence against a null model—not the probability that the null is true.

2. Intuition

Seeing 95 heads in 100 flips is very surprising if a coin is fair, so its p-value under fairness is tiny. That does not itself state the probability that the coin is unfair.

3. Prerequisites

Hypothesis testing, null distributions, test statistics, probability tails, significance thresholds, and sampling variability.

4. Core Concepts

Subtopic	Meaning	Why it matters / interview angle
Null conditionality	Calculated under H_0	Corrects the most common interpretation error.
Extremeness	At least as incompatible with H_0	Depends on statistic and alternative.
Threshold α	Prechosen decision boundary	.05 is policy, not a truth cutoff.
Effect size	Magnitude of difference	Tiny p can have tiny practical value.
Multiplicity	Many tests create chance small p-values	Requires adjustment or preselection.

5. Algorithm / Working Process

Choose null, alternative, statistic, tails, and α before data analysis. Compute observed statistic, find its null-tail probability, and report p-value with effect size and confidence interval. Make a predeclared decision, not a binary truth claim.

6. Mathematical Foundation

For statistic T , right-tailed p-value is $P(T \geq t_{\text{obs}} \mid H_0)$. A symmetric two-sided p-value is commonly $P(|T| \geq |t_{\text{obs}}| \mid H_0)$. Under true H_0 and valid assumptions, p-values are uniformly distributed on $[0, 1]$.

7. Practical Implementation

```
import numpy as np
from scipy.stats import ttest_1samp

latency_ms = np.array([102, 98, 101, 104, 99, 103, 100, 105])
result = ttest_1samp(latency_ms, popmean=100, alternative="two-sided")
print(f"mean diff={latency_ms.mean()-100:.2f} ms, t={result.statistic:.2f}, p={result.pvalue:.4f}")
```

8. Code Explanation

The test assumes a null population mean latency of 100 ms. Two-sided means deviations in either direction count. Mean difference communicates size; p-value communicates null-model compatibility.

9. Training / Evaluation

Do not declare one model better from a single accuracy difference alone. Preserve paired per-example predictions, avoid tuning repeatedly on the test set, and use a valid paired method. Report uncertainty and product relevance.

10. Complexity and Cost

Analytic calculation is usually $O(n)$. Permutation p-values cost $O(Bn)$ for B shuffles. Independent data collection is the real cost.

11. Common Use Cases

A/B-test dashboards; regression coefficient tests; feature experiments; model-improvement checks; scientific reporting.

12. Common Mistakes

Reading p as $P(H_0 \text{ true})$; treating .049 and .051 as meaningfully different; p-hacking; ignoring multiplicity, power, and effect size; choosing a one-sided test after seeing results.

13. Edge Cases / Limitations

Low power can yield high p for meaningful effects. Massive samples can yield tiny p for negligible effects. Dependence and invalid distribution assumptions produce misleading p-values.

14. Variations

One-/two-sided, exact, permutation, and adjusted p-values (Bonferroni, Benjamini–Hochberg). Bayesian posterior probabilities are a different answer, not p-values.

15. Related Topics

Hypothesis testing supplies decisions; confidence intervals provide effect ranges; power and sampling determine sensitivity; FDR handles multiple comparisons.

16. Interview Questions

1. **Define p-value.** Probability of equally/more extreme data assuming H_0 .
2. **Is it $P(H_0 \text{ true})$?** No.
3. **Does $p=.03$ prove usefulness?** No; inspect effect and cost.
4. **Does $p>.05$ prove no effect?** No; it is insufficient evidence at that α .
5. **Why set α first?** Prevents outcome-driven decisions.
6. **Why does n lower p ?** Larger n reduces standard error for a fixed effect.
7. **One/two-sided choice?** Predeclared directional vs non-directional claim.
8. **How handle 100 tests?** Control familywise error/FDR.
9. **What determines p ?** Effect, variance, n , and test design.
10. **What should accompany p ?** Effect estimate and CI.

17. Practice Tasks

Simulate fair coins; vary sample size for fixed effect; compare analytic and permutation p-values; apply Bonferroni and FDR corrections.

18. Project Ideas

Project	Stack and dataset	Resume value
Result explainer	Streamlit, SciPy	Prevents p-value misinterpretation.
Multiple-testing dashboard	Pandas, statsmodels	Demonstrates FDR control.
ML improvement validator	sklearn, NumPy	Tests paired prediction changes.

19. Quick Revision

Key idea: surprise under H_0 . Formula: null-tail probability. Use alongside effect/CI. Trap: it is not hypothesis probability. One-liner: p-values measure compatibility, not importance.

20. Final Cheat Sheet

Definition: null-conditional tail probability. **Input/output:** statistic/data $\rightarrow$ p-value. **Steps:** declare test, calculate statistic, use null tail. **Key parameters:** alternative, α , correction. **Pros:** familiar. **Cons:** easily misread. **Best use:** preplanned experiments.